

### 4.3 Maximum & Minimum Values

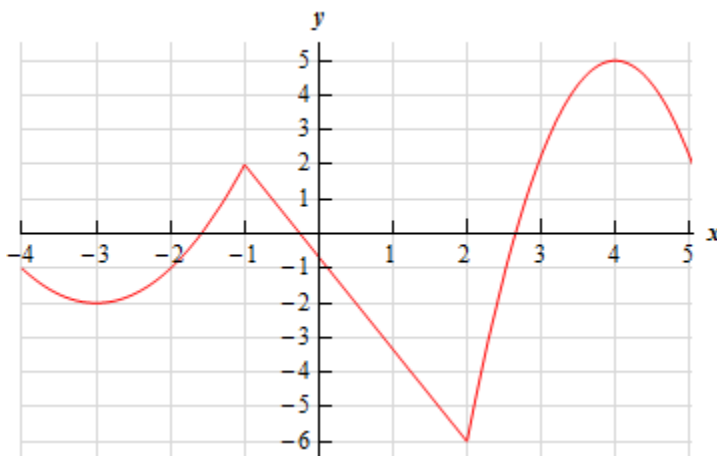
In the following sections we will begin to investigate how we can use derivatives to determine how various functions behave. In this section we are interested in determining where a function has its largest and smallest values. We define the following terms.

**Definition:** Suppose that  $c$  is a number in the domain  $D$  of a function  $f$ .

- We say that  $f(c)$  is an absolute maximum value of  $f$  on  $D$  if  $f(c) \geq f(x)$  for all  $x$  in  $D$ .
- We say that  $f(c)$  is an absolute minimum value of  $f$  on  $D$  if  $f(c) \leq f(x)$  for all  $x$  in  $D$ .
- We say that  $f(c)$  is a local maximum value of  $f$  on  $D$  if  $f(c) \geq f(x)$  when  $x$  is near  $c$ .
- We say that  $f(c)$  is a local minimum value of  $f$  on  $D$  if  $f(c) \leq f(x)$  when  $x$  is near  $c$ .

*Note: The precise meaning of the word “near” is that the inequality is true on some open interval containing  $c$ .*

**Example 1:** Identify where the graph below has local minimum and maximum values.



Local maximum of 2. Occurs at  $x = -1$ .

Local maximum of 5. Occurs at  $x = 4$ .

Local minimum of -2. Occurs at  $x = -3$ .

Local minimum of -6. Occurs at  $x = 2$ .

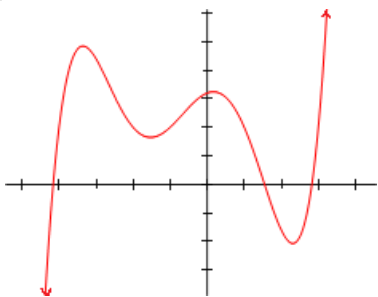
*Note: We cannot determine anything about the absolute min. & max. of this function since we cannot see the entire graph.*

**Example 2:** Identify the absolute max/min values of the following functions.

a)  $f(x) = \cos(x)$

Using knowledge from Precalculus we know that the largest value that  $f$  will have is 1 and the smallest value is -1. Therefore  $f(x)$  has an absolute maximum value of 1 and an absolute minimum value of -1. We also note that these values occur at infinitely many different locations.

b)  $g(x)$  (which is pictured below)



Here we can see that the graph does not have a largest or a smallest values. Therefore, this graph does not have an absolute maximum nor an absolute minimum. We note however that the graph does have several *local* minimums and maximums.

**Questions:** TRUE or FALSE

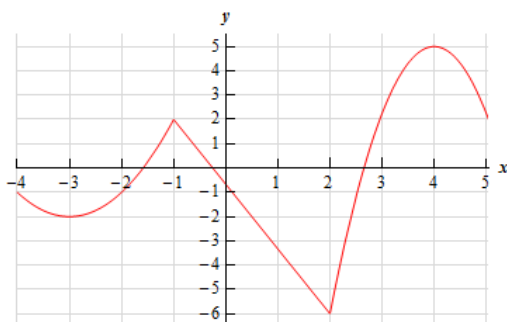
1) A function can have two absolute maximum values.

FALSE. If  $a$  and  $b$  were two different absolute maximum values then by definition  $a > b$  and  $b > a$  which is impossible.

2) A function's absolute maximum can equal its absolute minimum.

TRUE. Consider the function  $f(x) = 3$ . The absolute maximum and the absolute minimum of this function would be 3.

**Example 3:** Consider the function  $f(x)$  pictured below.



a) Determine  $f'(4)$ .

We want to know the slope of the tangent line to the curve at  $x = 4$ .  
Therefore we determine that  $f'(4) = 0$

b) Determine  $f'(-1)$ .

We want to know the slope of the tangent line to the curve at  $x = -1$ . Therefore we determine that  $f'(-1)$  DNE.

Intuitively, we discovered from this example that the first derivative of a function is either 0 or undefined at locations of local maximum/minimum values. The following theorem starts to make this idea precise.

**Theorem (Fermat's Theorem):** If  $f$  has a local minimum or maximum value at  $c$ , and if  $f'(c)$  exists then  $f'(c) = 0$ .

*Proof:* See section 4.3 of the textbook.

*Note:* Geometrically, Fermat's Theorem implies that there is a horizontal tangent at points corresponding to local max/min values on a graph (when the function is differentiable).

**Example 4:** Consider the function  $f(x) = x^3$ .

- a) Find the value(s) of  $x$  that make the first derivative of  $f$  equal to 0.

$$\text{We know that } f'(x) = 3x^2. \quad 3x^2 = 0 \Leftrightarrow x = 0$$

Therefore,  $f'(x) = 0$  only when  $x = 0$ .

- b) Is there a local min. or local max. of  $f$  at any of the values you found in part (a)?

By considering the graph of this function we realize that there is neither a min. nor a max. on the graph at  $x = 0$ .

- c) TRUE or FALSE: If  $g$  is a function and  $g'(c) = 0$ , then  $g$  must have a local min or local max at  $c$ .

This is FALSE. The counterexample is discovered in parts (a) and (b). In particular  $f'(0) = 0$ , yet there is not a local max nor a local min at 0.

*Note: Part (c) is quite important. It tells us that the converse of Fermat's Theorem does not hold. So, numbers which make the first derivative of a function equal to zero are only CANDIDATES for locations of local maxes/mins.*

Fermat's Theorem readily implies the following fact.

**Fact:** If  $f$  has a local maximum or minimum at  $c$ , then  $c$  is in the domain of  $f$  and either  $f'(c) = 0$  or  $f'(c)$  does not exist.

**Definition:** A critical number of a function  $f$  is a number  $c$  in the domain of  $f$  such that either  $f'(c) = 0$  or  $f'(c)$  does not exist.

*Note: The fact and definition above immediately imply that if  $f$  has a local max or min at  $c$ , then  $c$  is a critical number of  $f$ .*

**Example 5:** Find the critical numbers of  $f(x) = x^3 + 6x^2 - 15x$ .

First we note that  $f'(x) = 3x^2 + 12x - 15$ .

We notice that the first derivative of the function is never undefined.

So, the critical numbers must satisfy the equation:

$$f'(x) = 0 \Rightarrow 3x^2 + 12x - 15 = 0 \Rightarrow x^2 + 4x - 5 = 0 \Rightarrow (x+5)(x-1) = 0.$$

Thus, the critical numbers of the function are -5 and 1.

**Example 6:** Find all the critical numbers of  $f(x) = x^{3/5}(4-x)$ .

First, we note that by the product rule:

$$\begin{aligned} f'(x) &= \frac{3}{5}x^{-2/5}(4-x) + (-1)x^{3/5} \\ &= \frac{3(4-x)}{5x^{2/5}} - x^{3/5} \\ &= \frac{1}{x^{2/5}} \left[ \frac{3(4-x)}{5} - x \right] \\ &= \frac{1}{x^{2/5}} \left[ \frac{12-8x}{5} \right] \end{aligned}$$

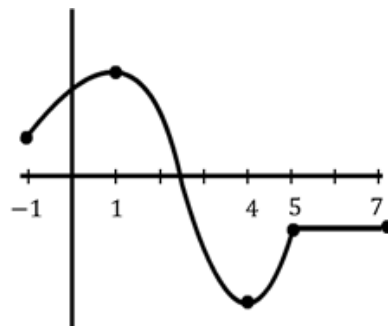
Clearly the first derivative is undefined when  $x$  is 0. Also

$$f'(x) = 0 \Rightarrow \frac{12-8x}{5} = 0 \Rightarrow 12-8x = 0 \Rightarrow 8x = 12 \Rightarrow x = \frac{3}{2}.$$

So, the critical numbers are 0 and 1.5.

**Question to the Class:** Is it possible to draw a picture of a graph of a continuous function on the closed interval  $[-1, 7]$  which does not have an absolute maximum value or does not have an absolute minimum value?

*Answer:* Intuitively this seems impossible if we have to keep the function continuous.



The next theorem makes the idea discussed above precise.

**Theorem (The Extreme Value Theorem):**

If  $f$  is continuous on a closed interval  $[a, b]$ , then  $f$  attains an absolute maximum value  $f(c)$  and an absolute minimum value  $f(d)$  at some numbers  $c$  and  $d$  in  $[a, b]$ .

*Note:* The extreme value theorem only tells us that absolute max/min values exist. It does not tell us how to find the values. The next procedure we go over will tell us how to actually find the absolute max/min values.

*Question:* Why would the extreme value theorem not hold on an open interval?

*Question:* Why would the extreme value theorem not hold if our function was not continuous?

**The Closed Interval Method:**

This method finds the absolute max/min values of a continuous function  $f$  on  $[a, b]$ .

1. Identify the critical points of  $f$  and the endpoints of the interval  $[a, b]$ .
2. Find the values of  $f$  at these values.
3. The largest  $y$ -value from step 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

*Note: For the closed interval method to work, we also have to be able to find the derivative of the function  $f$ .*

**Example 7:** Find the absolute extrema of  $f(x) = 2x^3 - 15x^2 + 36x$  on the interval  $[1, 5]$ .

Since  $f$  is continuous on  $[1, 5]$  it must have an absolute min. & max. on the interval  $[1, 5]$ .

We calculate that:  $f'(x) = 6x^2 - 30x + 36 = 6(x^2 - 5x + 6) = 6(x - 3)(x - 2)$

We see that  $f'(x)$  is never undefined and that  $f'(x) = 0$  when  $x = 2$  &  $x = 3$ . Therefore  $f$  has critical numbers of 2 & 3 (both of which are in the interval  $[1, 5]$ ).

So, our candidates for the absolute min/max occur at  $x = 2$  &  $x = 3$  and also at the endpoints of the interval  $x = 1$  &  $x = 5$ .

The values of  $f$  at our critical numbers and endpoints are:

$$f(1) = 23, \quad f(2) = 28, \quad f(3) = 27, \quad f(5) = 55$$

Therefore we see that  $f(x)$  has an absolute maximum of 55 and an absolute minimum of 23.

**Example 8:** Find the absolute extrema of  $f(x) = 6x^{4/3} - 3x^{1/3}$  on the interval  $[-1, 1]$ .

Since  $f$  is continuous on  $[-1, 1]$  it must have an absolute min. & max. on the interval  $[-1, 1]$ .

We calculate that:  $f'(x) = 8x^{1/3} - x^{-2/3} = x^{-2/3} [8x - 1] = \frac{8x - 1}{x^{2/3}}$

We see that  $f'(x)$  is undefined when  $x = 0$  and that  $f'(x) = 0$  when  $x = \frac{1}{8}$ . Therefore  $f$  has critical numbers of 0 &  $\frac{1}{8}$  (both of which are in the interval  $[-1, 1]$ ).

So, our candidates for the absolute min/max occur at  $x = 0$  &  $x = \frac{1}{8}$  and also at the endpoints of the interval  $x = -1$  &  $x = 1$ .

The values of  $f$  at our critical numbers and endpoints are:

$$f(-1) = 9, \quad f(0) = 0, \quad f\left(\frac{1}{8}\right) = -\frac{9}{8}, \quad f(1) = 3$$

Therefore we see that  $f(x)$  has an absolute maximum of 9 and an absolute minimum of  $-\frac{9}{8}$ .

**Example 9:** Find the exact absolute extrema of the function  $f(x) = x - 2\sin(x)$  on the interval  $[0, 2\pi]$ .

Since  $f$  is continuous on  $[0, 2\pi]$  it must have an absolute min. & max. on the interval  $[0, 2\pi]$ .

We calculate that:  $f'(x) = 1 - 2\cos(x)$

We see that  $f'(x)$  is never undefined and that  $f'(x) = 0$  when  $1 - 2\cos(x) = 0 \Leftrightarrow \cos(x) = \frac{1}{2}$ . In the interval  $[0, 2\pi]$  the only critical numbers of  $f$  are  $\frac{\pi}{3}$  and  $\frac{5\pi}{3}$ .

So, our candidates for the absolute min/max occur at  $x = \frac{\pi}{3}$  &  $x = \frac{5\pi}{3}$  and also at the endpoints of the interval  $x = 0$  &  $x = 2\pi$ .

The values of  $f$  at our critical numbers and endpoints are:

$$f(0) = 0, \quad f\left(\frac{\pi}{3}\right) = \frac{\pi}{3} - \sqrt{3} \approx -0.685, \quad f\left(\frac{5\pi}{3}\right) = \frac{5\pi}{3} + \sqrt{3} \approx 6.968, \quad f(2\pi) = 2\pi \approx 6.28$$

Therefore we see that  $f(x)$  has an absolute maximum of  $\frac{5\pi}{3} + \sqrt{3}$  and an absolute minimum of  $\frac{\pi}{3} - \sqrt{3}$ .